Count by hands

How a Hand Works

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little Wonderer counts fingers by fives, one hand at a time. What number comes next?

5, 10, 15, ___

Show your thinking.

2. Each thumb has 2 bones. Big Wonderer counts thumb bones by twos. What comes next?

4, 6, 8, ___

Show your thinking.

3. Two hands have 10 fingertips. They count fingertips by tens. What comes next?

30, 40, 50, ___

Show your thinking.

Count by hands

How a Hand Works

Name _____ Date _____

Show your thinking with numbers, marks, or words.

4. The little one counts string pulls in twos. What number is missing?

12, 14, ___, 18

Show your thinking.

5. They trace 5 strings on each hand outline. What number is missing?

25, ___, 35, 40

Show your thinking.

6. A jar holds elastic cords in bundles of 10. What number is missing?

50, 60, ___, 80

Show your thinking.

WONDER DAILY

How a Hand Works

A hand on strings

A mechanical hand toy stands open with five strings at the wrist. What inside a real hand works like it?

Bones in a row

Each finger has three small bones in a line, and the thumb has two. One whole hand holds 27 bones.

A string pulls

Forearm muscles pull long tendons, like strings. Tendons pull finger bones, and loops keep them close, like the toy's loops.

Where fingers bend

Joints are where bones meet. Finger joints fold like door hinges. The thumb's base joint lets it swing across the palm.

String to finger

Pull each toy string and match it to a finger. Then curl a real finger and feel the forearm move.

Opening again

Muscles only pull, never push. Tendons on the back of the hand pull fingers straight again. Toys may use elastic.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Explain a hand clue in your words

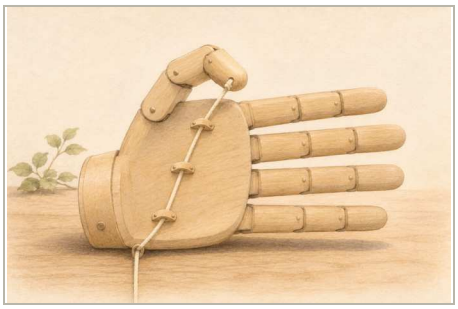
How a Hand Works · Curiosity day

Name _____ Date _____

Choose one picture. Say your idea first, then draw or write it.



Bones in a row



A string pulls



Opening again

Draw a hand outline. Mark the bones in one finger, one tendon, and the joints where it bends.

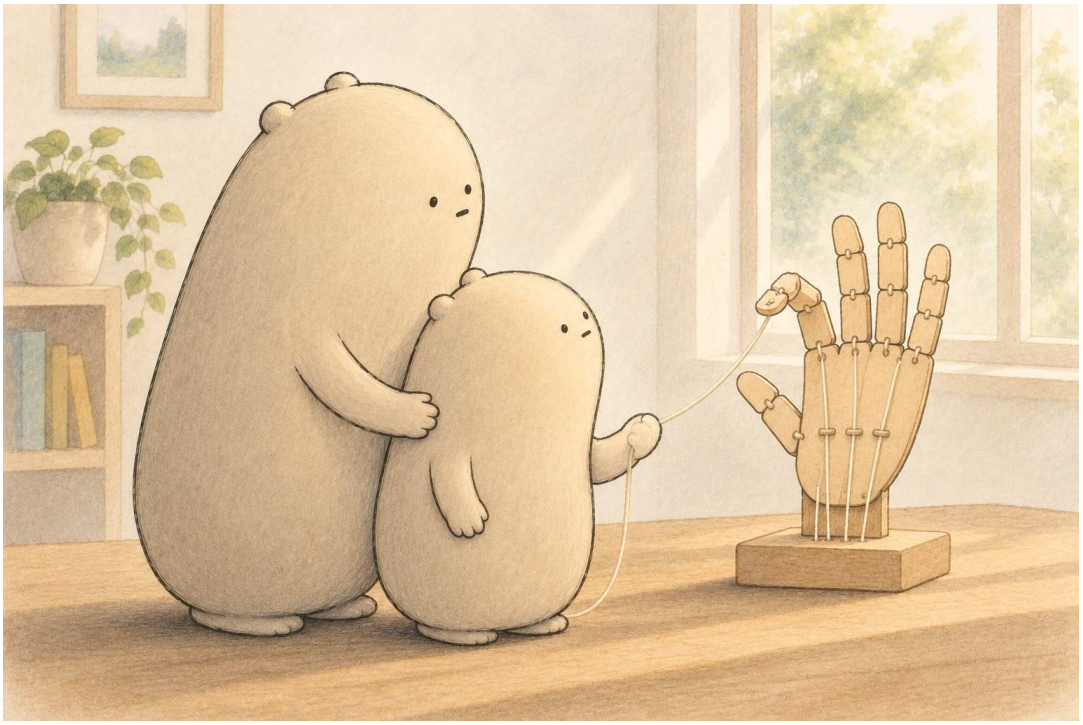
bone · tendon · joint · muscle · because · maybe

A finger stays stiff because … what is inside it?

A finger curls when … what pulls on it?

A finger opens again because … what can a muscle not do?

Does your sentence tell what you noticed and use because or maybe to explain?



Who Pulls My Strings?

Little Wonderer pulled on a string, only one,
and a toy finger curled like a cat in the sun.
"Who's pulling MY fingers?" the little one cried.
"I don't see a string, and I've looked on each side!"

Big Wonderer laughed. "You have strings, it is true.
They're tendons, tucked under your skin, out of view.
Your muscles sit up in your arm past your wrist,
and they tug, and they tug, till your fingers make fists."

Inside every finger are three bones in a row
(the thumb has just two, as the toy seemed to know),
with joints in between them that fold like a door —
just one way, not backward. (Please don't try for more.)

But muscles can't push, so to open back wide,
there are strings on the back, on the knuckle-y side.
So the little one waved at the toy on the shelf:
"We both work on strings — but I pull them myself!"

How a Hand Works

01 What keeps a finger stiff?



Stiff pieces in a row, like the bones in a finger.

Look closer. Find the short stiff pieces in each toy finger, and count them.

A little explanation. Bones. Each finger has three small bones in a row, and the thumb has two. Counting the wrist and palm, one hand has 27 bones. Bones hold the shape, but they cannot move by themselves.

Try this. Gently press along one of your fingers and count the hard parts between the bends. Count the stiff pieces on one toy finger too. Do the two numbers match?

02 What pulls a finger closed?



Pull a string, and one finger curls.

Look closer. Find the thin string that runs along the palm side of a curled toy finger.

A little explanation. In the toy, a string. In your hand, a tendon: a tough cord that joins a muscle to a bone. Muscles in your forearm pull long tendons that reach into each finger, and tough loops along the finger hold them close, like the toy's guides.

Try this. Draw around your hand on paper to make an outline. Pull one toy string slowly and watch which finger curls. Draw a line from that string to the same finger on your outline. Do the same for every string, one at a time.

03 Does your hand work like the toy?



Trace each string to the finger it curls.

Look closer. Watch where the toy finger bends, then watch your own finger bend.

A little explanation. Both follow one plan: a pull, a string, and joints that bend. Your finger bends at three joints, mostly one way, like a door hinge. The muscles that curl your fingers sit mostly in your forearm, so you can feel them move there.

Try this. Curl your pointer finger slowly and count the places it bends. Curl the toy's pointer finger and compare. Rest your other hand on your forearm and wiggle your fingers. What do you feel move?

04 What opens the hand again?



Something has to pull the finger open again.

Look closer. Look at the back of the toy hand for elastic, a spring, or a second string.

A little explanation. A muscle can only pull, never push. So a second set of tendons runs along the back of each finger and pulls it straight. Many toys use an elastic cord or a spring instead. Some tendons on the back of your hand are linked, which is why one finger can be hard to lift alone.

Try this. Let go of a toy string and watch whether the finger springs back. Look for what pulls it open. Then lay your hand flat, fold your middle finger under, and gently try to lift your ring finger.